```
#include<stdio.h>
int stack[100],n,top,i;
void push(int x)
{
    if(top>=n-1)
    {
        printf("\nSTACK is over flow");
    }
    else
    {
        top++;
        stack[top]=x;
        printf("\nelement pushed into stack:%d",x);
    }
}
void pop()
{
    if(top<=-1)
    {
        printf("\nStack is under flow");
    }
    else
    {
        printf("\nThe popped elements is %d",stack[top]);
        stack[top] = 0;
        top--;
    }
}
void display()
{
    if(top>=0)
    {
        printf("\n The elements in STACK: \n");
        for(i=top; i>=0; i--)
            printf("%d \n",stack[i]);
    }
    else
    {
        printf("\n The STACK is empty");
    }

}
int main()
{
      top = -1;
      n = 5;
      push(10);
      push(20);
      push(30);
      display();
      pop();
      display();
}
```